

MINI PROJECT - REPORT

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Title: CAR SALES – REPORT
Domain: Sales
Timeline of dataset: 2022-2025
Submission due date: Jan 8, 2026
Mentor name: Kumaran M

CAR SALES – REPORT

➤ Original Dataset Source

Source Link: [Raw Car sales Data set](#)

Data sets found in the link

1. Cars
2. Customers
3. Sales

Name	Date modified	Type	Size
 Cars.csv	01-01-2026 08:18	Microsoft Excel Co...	35 KB
 Customers.csv	01-01-2026 08:42	Microsoft Excel Co...	230 KB
 Sales.csv	01-01-2026 08:45	Microsoft Excel Co...	637 KB

1. Data Pre-processing using Excel

1.1 Blanks Identification

Found **Blanks** in the Brand Column of Cars dataset and updated with the Brand names based on the Models.

No Other blanks found in other files

Car_ID	Brand	Model	Year	Color	Engine_Type	Transmission
C0015		Altima	2019	Blue	Diesel	Manual
C0033		Model 3	2020	White	Petrol	Manual
C0086		5 Series	2019	Silver	Hybrid	Automatic
C0330		Tucson	2025	White	Diesel	Automatic

Car_ID	Brand	Model	Year	Color	Engine_Type
C0015	Nissan	Altima	2019	Blue	Diesel
C0033	Tesla	Model 3	2020	White	Petrol
C0086	BMW	5 Series	2019	Silver	Hybrid
C0330	Hyundai	Tucson	2025	White	Diesel

1.2 Duplicates Identification

Found and removed duplicates in Car ID of Cars data set.

No duplicates found in the other datasets

FileHomeInsertDrawPage LayoutFormulasDataReviewViewAutomateHelpAccount

▼fx=

Q0001

Car ID	Brand	Model	Year	Color	Engine_Type	Price	Quantity	Status
C0001	Toyota	Camry	2020	Red	Petrol	Automatic	60000.15	6 Available
Q0002	Tesla	Model 3	2019	Red	Electric	Manual	26437.73	16 Available
Q0003	Nissan	Qashqai	2018	Blue	Electric	Automatic	50156.13	20 Available
Q0004	Hyundai	Sonata	2020	Red	Hybrid	Auto	33026.14	3 Available
Q0005	Toyota	RAV4	2016	White	Hybrid	Manual	79672.9	9 Reserved
Q0006	Hyundai	Elantra	2019	White	Diesel	Automatic	94225.03	17 Sold
Q0007	Mercedes	C-Class	2020	Gray	Petrol	Manual	21244.38	7 Reserved
Q0008	Tesla	Model 3	2021	Gray	Hybrid	Manual	48813.34	14 Reserved
Q0009	BMW	3 Series	2017	Black	Petrol	Manual	48267.74	17 Reserved
Q0010	Tesla	Model S	2015	Red	Petrol	Manual	36005.2	10 Reserved
Q0011	Tesla	Model 3	2017	Red	Petrol	Manual	30555.84	2 Reserved
Q0012	Tesla	Model S	2016	Gray	Petrol	Manual	71104.74	15 Reserved
Q0013	Hyundai	Sonata	2018	Blue	Petrol	Manual	24426.04	2 Sold
Q0014	Toyota	RAV4	2024	Blue	Petrol	Manual	60670.75	2 Sold
Q0015	Altima	2019	Blue	Diesel	Manual	48267.74	17 Reserved	
Q0016	BMW	3 Series	2024	Black	Petrol	Manual	36005.2	10 Reserved
Q0017	Nissan	Qashqai	2018	Silver	Petrol	Automatic	30555.84	2 Reserved
Q0018	Nissan	Qashqai	2016	Black	Petrol	Manual	71104.74	15 Reserved
Q0019	Tesla	Model S	2019	White	Diesel	Manual	24426.04	2 Sold
Q0020	Toyota	Camry	2022	Blue	Electric	Manual	60670.75	2 Sold
Q0021	BMW	3 Series	2020	Red	Diesel	Automatic	60602.36	19 Available
Q0022	Tesla	Model X	2018	Red	Diesel	Automatic	40366.14	18 Sold
Q0023	Tesla	Model 3	2024	Silver	Diesel	Automatic	53377.41	20 Reserved
Q0024	Mercedes	E-Class	2019	Blue	Petrol	Manual	33346.72	8 Sold
Q0025	Nissan	Altima	2016	Silver	Petrol	Manual	48204.7	2 Available

Microsoft Excel

1 duplicate rows found and removed. 100 unique values remain. Note that counts may include empty cells, spaces, etc.

OK

1.3 Find and Replace

Found inconsistent data in the Transmission column of the cars dataset and used find and replace to correct the same.

Car_ID	Brand	Model	Year	Color	Engine_Type	Transmission	Price	Quantity_In_Stock	Status
C0098	Tesla	Model X	2017	Blue	Electric	Automatic	99875.13	17	Reserved
C0390	Nissan	Sunny	2023	Blue	Petrol	Automatic	39960.25	1	Reserved
C0400	Toyota	Corolla	2016	Gray	Petrol	Automatic	97026.32	14	Reserved
C0420	Tesla	Model X	2021	Black	Diesel	Automatic	99307.62	5	Reserved
C0430	Nissan	Sunny	2017	Red	Diesel	Automatic	21393.33	6	Reserved
C0491	Kia	Seltos	2022	Red	Petrol	Automatic	19841.22	5	Sold

Microsoft Excel

All done. We made 6 replacements.

Find and Replace

Find what: auto

Replace with: Automatic

Replace All

Car_ID	Brand	Model	Year	Color	Engine_Type	Transmission	Price	Quantity_In_Stock	Status
C0950	Mercedes	C-Class	2024	Red	Diesel	Manual	88724.54	16	Reserved
C0343	Nissan	Sunny	2019	Red	Hybrid	Manual	83159.71	19	Sold
C0348	Hyundai	Tucson	2019	Red	Electric	Manual	16688.91	1	Reserved

Microsoft Excel

All done. We made 3 replacement

Find and Replace

Find what: M

Replace with: Manual

Replace All

Car_ID	Brand	Model	Year	Color	Engine_Type	Transmission	Price	Quantity_In_Stock	Status
C0371	Hyundai	Tucson	2023	Red	Hybrid	A	34068.8	14	Reserved

Find and Replace

Find what: A

Replace with: Automatic

Find Next

Car_ID	Brand	Model	Year	Color	Engine_Type	Transmission	Price	Quantity_In_Stock	Status
C0485	Tesla	Model X	2022	Gray	Petrol	Manual	95460.71	8	Sold

Find and Replace

Find what: manual

Replace with: Manual

Replace

1.4 Proper

Used Proper to correct the Lastname in the Customers dataset

File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat Table Design												
C2		=PROPER([@[Last Name]])										
Customer_ID	First Name	Last Name	Last Name	Gender	Age	Job Role	Phone	Email	City	State	Region	
CU0001	Jill	snyder	=PROPER([@[Last Name]])			Network Eng	1-202-347-timothy92	Oakterr	Alabama	South		
CU0002	Nicholas	foster		Male	23	Librarian	1-752-982-cochranc	West Clea	Alaska	West		
CU0003	Courtney	robbins		Male	60	Surgeon	1-774-292-donna01	Mount Shr	Arizona	West		
CU0004	Blake	barry		Male	67	Inventory M	1-553-463-sandra08	North Sho	Arkansas	South		
CU0005	Claudia	hardin		Female	31	Librarian	1-218-426-caitlindav	Saint Oak	California	West		
CU0006	Scott	wilson		Female	52	Occupation	1-536-274-pamela97	Mount Lak	Colorado	West		
CU0007	Michael	johnson		Male	58	UI/UX Desig	1-685-543-rjones@b	New Sprin	Connectic	Northeast		
CU0008	Michelle	davis		Female	28	Machine Op	1-753-971-dwilson@	Sunsetrid	Delaware	South		
CU0009	Sara	mack		Male	29	Project Cool	1-805-949-ybailey@y	Port Glenr	Florida	South		
CU0010	Ashley	butler		Female	30	Process Eng	1-460-414-williames	Port Lakes	Georgia	South		
CU0011	Charles	sullivan		Male	65	Welder	1-313-561-fward@pi	North Cop	Hawaii	West		
CU0012	Tiffany	thompson		Male	30	Network Eng	1-449-893-lindseychi	Lake Rock	Idaho	West		
CU0013	Juan	blackwell		Male	23	Financial Ar	1-789-938-dguerrero	Saint Redh	Illinois	Midwest		
CU0014	Todd	taylor		Female	53	Anesthesiol	1-794-921-rodriguez	South Pink	Indiana	Midwest		
CU0015	Brooke	bennett		Male	23	CNC Machir	1-809-903-marvin28	New Rock	Iowa	Midwest		
CU0016	Bobby	rod%riguez		Female	68	Process Eng	1-582-511-torresalle	Whisperri	Kansas	Midwest		
CU0017	Patricia	french		Female	60	Academic A	1-730-892-sheila92	Port Brigh	Kentucky	South		
CU0018	Nancy	reed		Male	25	Accountant	1-293-343-jenningsfc	New Gran	Louisiana	South		
CU0019	Timothy	valenzuela		Female	42	Software En	1-232-732-jenkinsmi	Sapphirep	Maine	Northeast		
CU0020	Gregory	carter		Male	35	Stock Clerk	1-416-489-clayaman	West Brigh	Maryland	South		
CU0021	Diane	glenn		Male	32	Surgeon	1-558-690-bergrenee	Mount Rid	Massachu	Northeast		
CU0022	Jasmine	dixon		Male	58	Plant Manaj	1-611-718-sjefferson	North Sun	Michigan	Midwest		
CU0023	Anna	reynolds		Female	25	Solutions Ar	1-984-874-emilyfry@	Havenridg	Minnesot	Midwest		
CU0024	Timothy	delgado		Female	56	Data Scienti	1-219-340-robertdou	Saint Sierr	Mississip	South		
CU0025	Troy	hernandez		Male	52	Production	1-220-749-maynandr	Riveraook	Missouri	Midwest		

1.5 Concatenate

Used concatenate to merge the first name and last name columns and create a new column as Full Name

D3		=CONCATENATE([@[First Name]], " ", [@[Last Name2]])										
Customer_ID	First Name	Last Name	Full name	Gender	Age	Job Role	Phone	Email	City	State	Region	
CU0001	Jill	Snyder	Jill Snyder	Male	49	Network Eng	1-202-347-timothy92	Oakterr	Alabama	South		
CU0002	Nicholas	Foster	Nicholas Foster	Male	23	Librarian	1-752-982-cochranc	West Clea	Alaska	West		
CU0003	Courtney	Robbins	Courtney Robt	Male	60	Surgeon	1-774-292-donna01	Mount Shr	Arizona	West		
CU0004	Blake	Barry	Blake Barry	Male	67	Inventory M	1-553-463-sandra08	North Sho	Arkansas	South		
CU0005	Claudia	Hardin	Claudia Hardi	Female	31	Librarian	1-218-426-caitlindav	Saint Oak	California	West		
CU0006	Scott	Wilson	Scott Wilson	Female	52	Occupation	1-536-274-pamela97	Mount Lak	Colorado	West		
CU0007	Michael	Johnson	Michael Johns	Male	58	UI/UX Desig	1-685-543-rjones@b	New Sprin	Connectic	Northeast		
CU0008	Michelle	Davis	Michelle Davi	Female	28	Machine Op	1-753-971-dwilson@	Sunsetrid	Delaware	South		
CU0009	Sara	Mack	Sara Mack	Male	29	Project Cool	1-805-949-ybailey@y	Port Glenr	Florida	South		
CU0010	Ashley	Butler	Ashley Butler	Female	30	Process Eng	1-460-414-williames	Port Lakes	Georgia	South		
CU0011	Charles	Sullivan	Charles Sulli	Male	65	Welder	1-313-561-fward@pi	North Cop	Hawaii	West		
CU0012	Tiffany	Thompson	Tiffany Thomp	Male	30	Network Eng	1-449-893-lindseychi	Lake Rock	Idaho	West		
CU0013	Juan	Blackwell	Juan Blackwel	Male	23	Financial Ar	1-789-938-dguerrero	Saint Redh	Illinois	Midwest		
CU0014	Todd	Taylor	Todd Taylor	Female	53	Anesthesiol	1-794-921-rodriguez	South Pink	Indiana	Midwest		
CU0015	Brooke	Bennett	Brooke Benne	Male	23	CNC Machir	1-809-903-marvin28	New Rock	Iowa	Midwest		
CU0016	Bobby	Rod%Riguez	Bobby Rod%R	Female	68	Process Eng	1-582-511-torresalle	Whisperri	Kansas	Midwest		
CU0017	Patricia	French	Patricia Frenc	Female	60	Academic A	1-730-892-sheila92	Port Brigh	Kentucky	South		
CU0018	Nancy	Reed	Nancy Reed	Male	25	Accountant	1-293-343-jenningsfc	New Gran	Louisiana	South		
CU0019	Timothy	Valenzuela	Timothy Valer	Female	42	Software En	1-232-732-jenkinsmi	Sapphirep	Maine	Northeast		
CU0020	Gregory	Carter	Gregory Carter	Male	35	Stock Clerk	1-416-489-clayaman	West Brigh	Maryland	South		
CU0021	Diane	Glenn	Diane Glenn	Male	32	Surgeon	1-558-690-bergrenee	Mount Rid	Massachu	Northeast		
CU0022	Jasmine	Dixon	Jasmine Dixor	Male	58	Plant Manaj	1-611-718-sjefferson	North Sun	Michigan	Midwest		
CU0023	Anna	Reynolds	Anna Reynold	Female	25	Solutions Ar	1-984-874-emilyfry@	Havenridg	Minnesot	Midwest		
CU0024	Timothy	Delgado	Timothy Delga	Female	56	Data Scienti	1-219-340-robertdou	Saint Sierr	Mississip	South		

1.6 Pivot Table

Created Pivot table to identify number of cars sold on each Car Id and Number of sales done by salesperson.

Row Labels	Count of Sale_ID	Row Labels	Sum of Quantity
Aaron Allen	1	C0001	35
Aaron Austin	1	C0002	36
Aaron Brown	1	C0003	54
Aaron Carroll	1	C0004	19
Aaron Casey	1	C0005	35
Aaron Davis	1	C0006	37
Aaron Dudley	1	C0007	47
Aaron Farrell	1	C0008	38
Aaron Fitzgerald	1	C0009	38
Aaron Gill	1	C0010	34
Aaron Greer	1	C0011	41
Aaron Hamilton	1	C0012	44
Aaron Hansen	1	C0013	35
Aaron Hendrix	1	C0014	35
Aaron Johnson	1	C0015	59
Aaron Kennedy	1	C0016	41
Aaron Mcgee	1	C0017	43
Aaron Mcguire	1	C0018	39
Aaron Mitchell	1	C0019	58
Aaron Osborne	1	C0020	48

1.7 Data Analysis

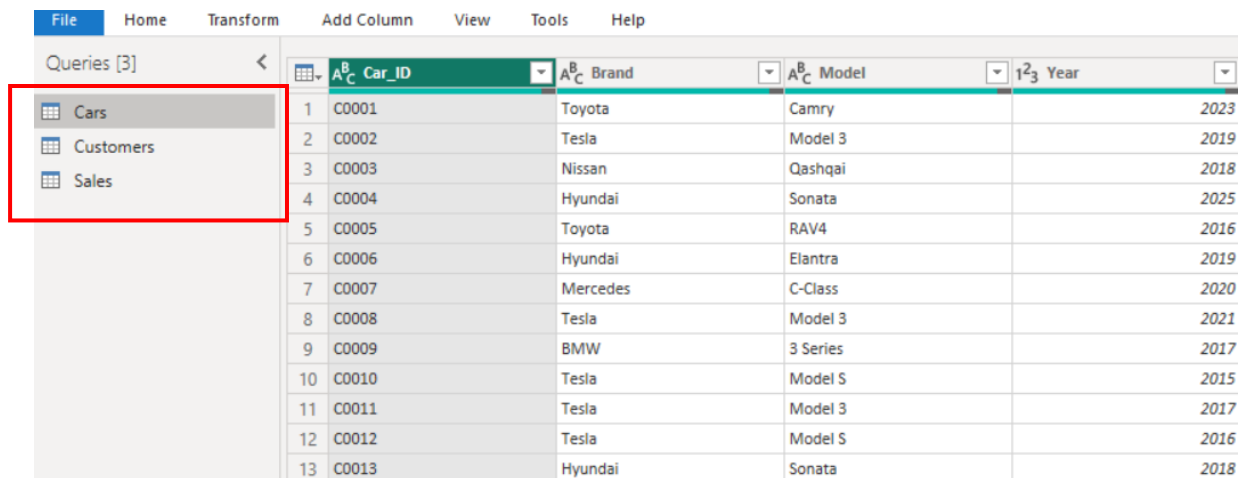
Used Data Analysis tool to identify the sales statistics.

Sales statistics	
Mean	57421.35441
Standard Error	245.7008235
Median	56934.485
Mode	69039.52
Standard Deviation	24570.08235
Sample Variance	603688946.7
Kurtosis	-1.196340416
Skewness	0.015088904
Range	84994.85
Minimum	15000.37
Maximum	99995.22
Sum	574213544.1
Count	10000
Largest(1)	99995.22
Smallest(1)	15000.37
Confidence Level(95.0%)	481.6230647

2. Data Visualization using Power BI

2.1 Load Data in the Power BI

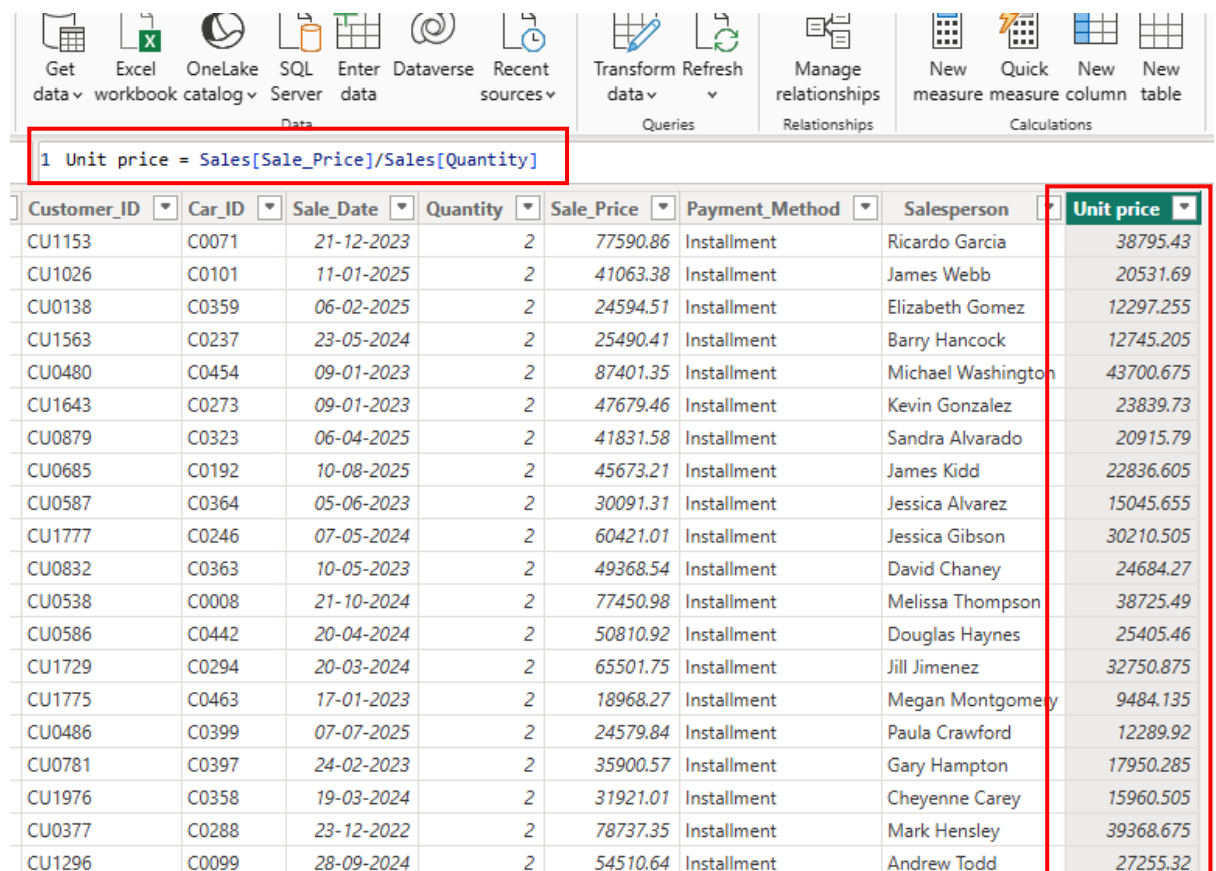
Loaded the excel files in the Power BI



Car_ID	Brand	Model	Year
1 C0001	Toyota	Camry	2023
2 C0002	Tesla	Model 3	2019
3 C0003	Nissan	Qashqai	2018
4 C0004	Hyundai	Sonata	2025
5 C0005	Toyota	RAV4	2016
6 C0006	Hyundai	Elantra	2019
7 C0007	Mercedes	C-Class	2020
8 C0008	Tesla	Model 3	2021
9 C0009	BMW	3 Series	2017
10 C0010	Tesla	Model S	2015
11 C0011	Tesla	Model 3	2017
12 C0012	Tesla	Model S	2016
13 C0013	Hyundai	Sonata	2018

2.2 Create New column

Created a new calculated column for finding the Unit price



1 Unit price = Sales[Sale_Price]/Sales[Quantity]

Customer_ID	Car_ID	Sale_Date	Quantity	Sale_Price	Payment_Method	Salesperson	Unit price
CU1153	C0071	21-12-2023	2	77590.86	Installment	Ricardo Garcia	38795.43
CU1026	C0101	11-01-2025	2	41063.38	Installment	James Webb	20531.69
CU0138	C0359	06-02-2025	2	24594.51	Installment	Elizabeth Gomez	12297.255
CU1563	C0237	23-05-2024	2	25490.41	Installment	Barry Hancock	12745.205
CU0480	C0454	09-01-2023	2	87401.35	Installment	Michael Washington	43700.675
CU1643	C0273	09-01-2023	2	47679.46	Installment	Kevin Gonzalez	23839.73
CU0879	C0323	06-04-2025	2	41831.58	Installment	Sandra Alvarado	20915.79
CU0685	C0192	10-08-2025	2	45673.21	Installment	James Kidd	22836.605
CU0587	C0364	05-06-2023	2	30091.31	Installment	Jessica Alvarez	15045.655
CU1777	C0246	07-05-2024	2	60421.01	Installment	Jessica Gibson	30210.505
CU0832	C0363	10-05-2023	2	49368.54	Installment	David Chaney	24684.27
CU0538	C0008	21-10-2024	2	77450.98	Installment	Melissa Thompson	38725.49
CU0586	C0442	20-04-2024	2	50810.92	Installment	Douglas Haynes	25405.46
CU1729	C0294	20-03-2024	2	65501.75	Installment	Jill Jimenez	32750.875
CU1775	C0463	17-01-2023	2	18968.27	Installment	Megan Montgomery	9484.135
CU0486	C0399	07-07-2025	2	24579.84	Installment	Paula Crawford	12289.92
CU0781	C0397	24-02-2023	2	35900.57	Installment	Gary Hampton	17950.285
CU1976	C0358	19-03-2024	2	31921.01	Installment	Cheyenne Carey	15960.505
CU0377	C0288	23-12-2022	2	78737.35	Installment	Mark Hensley	39368.675
CU1296	C0099	28-09-2024	2	54510.64	Installment	Andrew Todd	27255.32

2.3 Change Column format

Changed the column format of Unit price to Whole number

Name

Price

Format

Whole number

Summarization

Sum

Data category

Uncategorized

Sort by column

Sort

Data groups

Groups

Structure

Formatting

Properties

Car_ID	Brand	Model	Year	Color	Engine_Type	Transmission	Price	Quantity_In_Stock	Status
CO007	Mercedes	C-Class	2020	Gray	Petrol	Manual	21,344	7	Reserved
CO016	BMW	3 Series	2024	Black	Petrol	Manual	36,006	10	Reserved
CO018	Nissan	Qashqai	2016	Black	Petrol	Manual	71,105	15	Reserved
CO024	Mercedes	E-Class	2019	Blue	Petrol	Manual	33,547	8	Sold
CO025	Nissan	Altima	2016	Silver	Petrol	Manual	48,205	2	Available
CO033	Tesla	Model 3	2020	White	Petrol	Manual	21,841	10	Sold
CO059	Kia	Sportage	2020	Gray	Petrol	Manual	91,552	4	Sold
CO061	Nissan	Sunny	2019	Black	Petrol	Manual	70,325	19	Available
CO081	Tesla	Model 3	2016	Black	Petrol	Manual	37,385	12	Reserved
CO084	BMW	3 Series	2024	White	Petrol	Manual	45,428	18	Available

2.4 Create New Measures

Created a New Measure to find YTD sales data, Total Sales Value

Name

Sales YTD

Format

Whole number

Data category

Uncategorized

New measure

Quick measure

Calculations

Home table

Cars

Structure

Formatting

\$

%

0

Properties

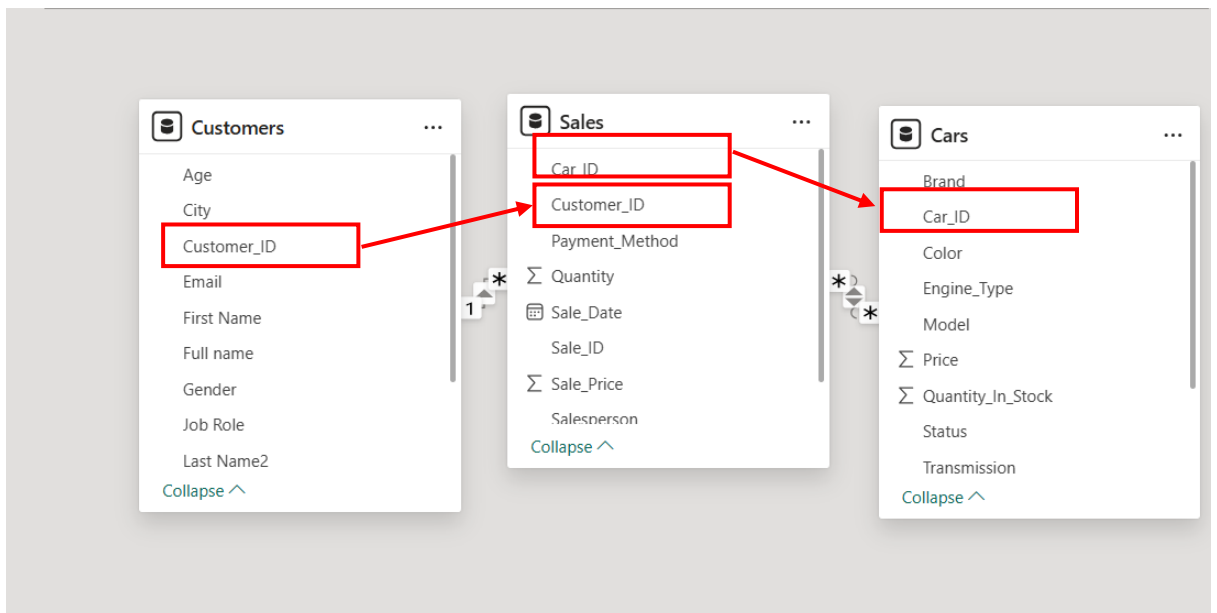
1 Sales YTD = CALCULATE(sum(Sales[Sales_Price]),DATESYTD(Sales[Sales_Date].[Year]))

- ▼  Cars
 - ☐ Brand
 - ☐ Car_ID
 - ☐ Color
 - ☐ Engine_Type
 - ☐ Model
 - ☐ Σ Price
 - ☐ Σ Quantity_In_Stock
 - ☐  Sales YTD
 - ☐ Status
 - ☐  Total sales Value
 - ☐ Transmission
 - ☐ Σ Year
- >  Customers
- ▼  Sales
 - ☐ Car_ID
 - ☐ Customer_ID
 - ☐ Payment_Method
 - ☐ Σ Quantity
 - > ☐  Sale_Date
 - ☐ Sale_ID
 - ☐ Σ Sale_Price
 - ☐ Salesperson
 - ☐  Unit price

2.5 Relationship

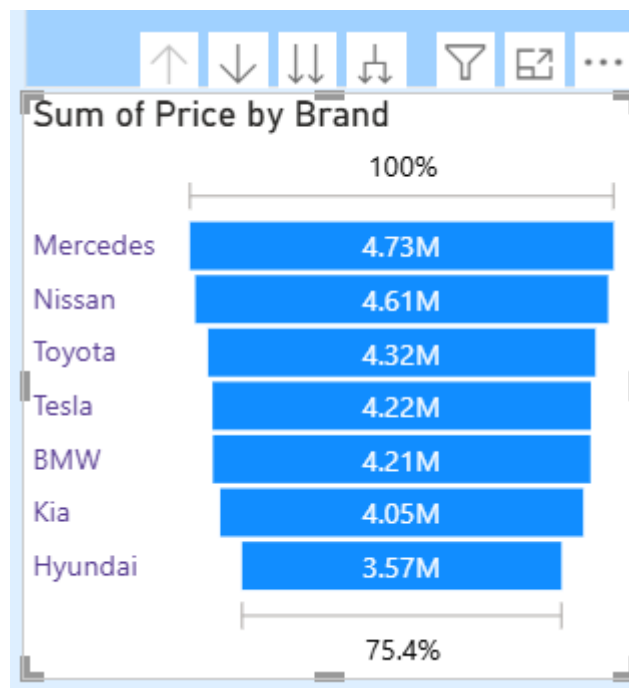
Created relationships between the three tables in manage relationship.

Linked Customer Id from Customers to Sales and Car ID from Cars to Sales.



2.6 Drill Through

Created Drill Through in the funnel chart.



2.7 Dashboard

Created a Dashboard using

1. **Bar Chart** – Average unit price by Brand and model
2. **Slicers** – Used Sales date as slicer to filter by date range
3. **Donut Chart** – Used Donut chart to identify sales by region
4. **Clustered Columns** – Used Clustered column to map Engine types sold each year
5. **Cards** – Used Cards to Highlight Total sales value and total number of sales
6. **Stacked Column Chart** - Used column chart to visualize total sales by payment method
7. **Funnel Chart** – Used Funnel chart to visualize Average price by each brand
8. **Table** – Used table to visualize stock quantity available based on brand
9. **Pie Chart** – Used Pie chart to visualize the number of models available in Each Brands
10. **Map** – Used map Chart to visualize the spread of sales at state.

Aligned all charts to enhance the dashboard's appearance and updated labels and titles for better clarity.

